

Detailed Theory: Ethernet

1. Overview of Ethernet

Ethernet is the most widely used technology for Local Area Networks (LANs). It defines a set of standards for physical wiring, signaling, frame formats, and protocols to enable devices on a LAN to communicate.

- **Origin:** Developed in the 1970s by Xerox PARC, later standardized by IEEE as IEEE 802.3.
- **Purpose:** To enable reliable data transmission over a shared medium.
- **Scope:** Primarily used for wired LANs.

2. Ethernet Standards and Evolution

- **10BASE-T:** 10 Mbps, baseband signaling over twisted-pair cables (Cat3 or better), max length 100 meters.

- **100BASE-TX (Fast Ethernet):** 100 Mbps over twisted-pair Cat5 cables.
 - **1000BASE-T (Gigabit Ethernet):** 1 Gbps over Cat5e or Cat6.
 - **10GBASE-T:** 10 Gbps over twisted-pair copper cables (Cat6a/Cat7).
 - **Other variants:** Fiber optic standards like 1000BASE-SX, 10GBASE-SR.
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3. Ethernet Frame Structure

Ethernet frames follow a strict format to encapsulate data for transmission:

Field	Size (Bytes)	Description
Preamble	7	Synchronizes receiver clocks
Start Frame Delimiter (SFD)	1	Marks start of frame

Destination MAC	6	MAC address of recipient
Source MAC	6	MAC address of sender
EtherType/Length	2	Protocol type or length field
Payload/Data	46 - 1500	Actual data being transmitted (minimum 46 bytes to ensure frame size)
Frame Check Sequence (FCS)	4	CRC for error checking

4. MAC Addresses

- **Length:** 48 bits (6 bytes).
- **Format:** Usually represented in hexadecimal, e.g., 00:1A:2B:3C:4D:5E.
- **Uniqueness:** Globally unique, assigned by manufacturers.
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Types:

- Unicast: Single recipient.
- Multicast: Group of devices.
- Broadcast: All devices on LAN (FF:FF:FF:FF:FF:FF).

5. Media Access Control (MAC) Protocols in Ethernet

- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):**
Early Ethernet networks used CSMA/CD to avoid collisions by sensing the medium before transmitting and backing off if collisions occur.
- **How CSMA/CD Works:**
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Sense the carrier (cable).

2.

If idle, transmit data.

3.

If collision detected, send jam signal and wait random time before retrying.

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Limitations:

CSMA/CD is obsolete in modern switched Ethernet, where full-duplex links prevent collisions.

6. Switches and Hubs in Ethernet

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Hubs: Broadcast incoming packets to all ports, causing collisions.

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Switches: Forward frames only to the destination port using MAC address tables, eliminating collisions and increasing efficiency.

7. Full Duplex vs Half Duplex

- **Half Duplex:** Transmission occurs in both directions, but not simultaneously (CSMA/CD used).
- **Full Duplex:** Transmission can occur simultaneously in both directions; no collisions.

8. Ethernet Physical Media

- **Twisted-Pair Copper Cables:** Most common (Cat5, Cat6).
- **Fiber Optic:** Used for longer distances and higher speeds.
- **Coaxial Cable:** Early Ethernet (10BASE2, 10BASE5), now obsolete.

9. VLANs (Virtual LANs)

- VLANs allow segmenting a physical Ethernet network into multiple logical networks.
- Uses 802.1Q tagging to insert VLAN ID in Ethernet frames.
- Enhances security and reduces broadcast domains.

10. Ethernet in Modern Networks

- Primarily uses switches instead of hubs.
- Supports Power over Ethernet (PoE) to supply power to devices.
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Backbone for LANs and many WAN access points.

Detailed Theory: Wireless Networking

1. Overview of Wireless Networking

Wireless Networking allows devices to connect and communicate without physical cables, using radio waves or infrared signals.

- **Key benefit:** Mobility and ease of installation.
 - **Types:** WLAN (Wi-Fi), WPAN (Bluetooth), WMAN (WiMAX), WWAN (cellular networks).
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2. Wireless Standards: IEEE 802.11 Family

- **802.11a:** 5 GHz, up to 54 Mbps.

- 802.11b: 2.4 GHz, up to 11 Mbps.
- 802.11g: 2.4 GHz, up to 54 Mbps.
- 802.11n: 2.4/5 GHz, up to 600 Mbps using MIMO.
- 802.11ac: 5 GHz, up to several Gbps.
- 802.11ax (Wi-Fi 6): Improved efficiency, throughput, and capacity.

3. Wireless Network Architecture

- Basic Service Set (BSS):
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Single access point (AP) with clients connected.

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Extended Service Set (ESS):

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Multiple BSS interconnected for wider coverage.

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Ad Hoc Mode: Devices connect directly without AP.

4. Wireless Communication Fundamentals

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Frequency Bands: 2.4 GHz and 5 GHz are most common.

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Channels: Divisions within frequency bands to avoid interference.

- **Modulation:** Techniques like OFDM to encode data on radio waves.
 - **MIMO (Multiple Input Multiple Output):** Uses multiple antennas to increase throughput and reliability.
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5. Wireless Security Protocols

- **WEP (Wired Equivalent Privacy):** Early, weak security.
 - **WPA (Wi-Fi Protected Access):** Improved security.
 - **WPA2:** Strong encryption with AES.
 - **WPA3:** Latest, stronger security and protections.
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6. Wireless MAC Layer

- Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) because collisions can't be detected easily in wireless.
 - RTS/CTS (Request to Send / Clear to Send): Mechanism to reduce collisions by reserving the medium before sending.
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7. Wireless Challenges

- **Interference:** From other wireless devices, microwaves, etc.
- **Signal attenuation:** Signal weakens over distance and obstacles.
- **Multipath fading:** Reflections cause signals to interfere destructively.
- **Security threats:** Eavesdropping, rogue APs, and man-in-the-middle attacks.

8. Wireless Devices

- **Access Point (AP):** Connects wireless clients to wired networks.
- **Wireless Router:** Combines router and AP.
- **Client devices:** Smartphones, laptops, IoT devices.

9. Wireless Network Topologies

- **Infrastructure mode:** Clients connect through AP.
- **Ad hoc mode:** Peer-to-peer connections without AP.

- **Mesh networks:** Devices act as routers to extend coverage dynamically.
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10. Recent Advances

- **Wi-Fi 6E:** Expands Wi-Fi 6 into 6 GHz band.
- **MU-MIMO:** Multi-user MIMO for simultaneous communication.
- **Beamforming:** Focused signal direction to improve range and throughput.